

Random Structure Search

Top 4:

Here we see how much can be explored in 100 steps with $\sigma=10$, changing only the temperature.

With higher T, more steps with an increase in energy are accepted in the acceptance criterion, because we divide by a larger number.

Example:

$$\exp(-1/0.1) = 0.000045$$

$$\exp(-1/2) = 0.60$$

Bottom 4:

Here we see how much can be explored in 100 steps with $kT=0.5$, changing only the width of the normal distribution for x' .

For the smaller sigma, the steps on the x-axis are small.

With larger sigma, the steps on the x-axis are bigger.

Conclusion:

One could assume that a smaller temperature $kT < 0.5$ with a wider distribution $\sigma > 10$ yields a faster convergence rate.

FYI: All start out at $x=3$. The red dot records the state with the lowest energy that was visited.

